UNIT 1 · 9 hours

Generation of Sinusoidal Voltage



Circuit Fundamentals

Basic Electrical and Electronics Engineering (EEE103)
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NMAM Institute of Technology, Nitte · 2025-26

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Learning objectives

- Explain how a rotating coil generates a sinusoidal EMF
- Define cycle, period, frequency, amplitude, phase
- Write the instantaneous equation $e = E_m \sin \omega t$
- Relate ω, f and T

Syllabus context

Basic nodal and mesh analysis excited by independent DC voltage sources, Power and Energy. Generation of sinusoidal voltage, frequency, average value, RMS value, form factor and peak factor, phasor representation of alternating quantities.

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Why alternating current?

Easy to generate at high voltage by rotating machines.

Transformers step voltage up for transmission and down for use, impossible with DC.

Motors such as the induction motor, run directly from AC.

Standard supply: 230 V (RMS), 50 Hz single-phase in India.

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Generating an EMF: coil in a field

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Coil

A coil AB rotates at uniform speed in a uniform magnetic field.

2

Flux cutting

Sides of the coil cut flux; Faraday's law gives EMF e = B l v sin θ.

3

Zero EMF

When the coil sides move parallel to the field (position 1), no flux is cut.

4

Peak EMF

When sides move perpendicular to the field (position 3), cutting is maximum.

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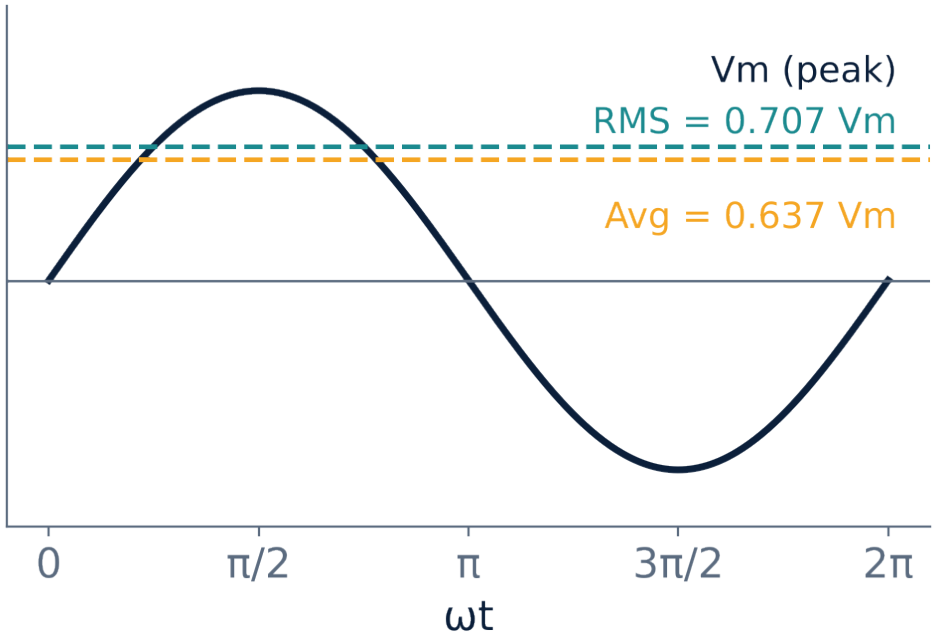
Reversal

In the second half-cycle the polarity reverses; the cycle repeats.

One revolution of a two-pole machine = one complete cycle of EMF.

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The sinusoidal waveform



- e = Em sin ωt = Em sin θ
- θ = ωt (radians)
- ω = 2πf = 2π/T rad/s
- f = 1/T (hertz)
- 50 Hz → T = 20 ms, ω = 314 rad/s

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Terms used with alternating quantities

Term	Meaning
Waveform	Graph of the quantity against time or angle
Cycle	One complete repetition of the variation
Time period T	Duration of one cycle (s)
Frequency f	Number of cycles per second (Hz) = 1/T
Instantaneous value	Value at any instant: e, v, i
Peak (amplitude) Em	Maximum instantaneous value
Peak-to-peak	Maximum positive to maximum negative: 2Em for a sine
Phase	Fraction of the cycle advanced from the reference

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Example: reading an equation

Problem An alternating voltage is v = 141.4 sin 377t volts. Find peak value, frequency, period and v at t = 3 ms.

- Peak Vm = 141.4 V
- f = ω/2π = 377/2π = 60 Hz, T = 1/60 = 16.67 ms
- v(3 ms) = 141.4 sin(377 × 0.003 rad) = 141.4 sin(1.131 rad = 64.8°)

Answer: Vm = 141.4 V, f = 60 Hz, v = 127.94 V

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Frequency of a multi-pole generator

- One pair of poles → one cycle per revolution.
- P poles, speed N rpm: $f = P N / 120$ Hz.
- Example: a 4-pole alternator at 1500 rpm gives $f = 4 \times 1500 / 120 = 50$ Hz.
- This is the same relation as the synchronous speed of an induction motor, $N_s = 120 f / P$.

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Key takeaways

- A coil rotating in a magnetic field induces $e = E_m \sin \omega t$
- $\omega = 2\pi f$ and $f = 1/T$
- Peak-to-peak value of a sine is twice the peak
- $f = PN/120$ for a P-pole machine at N rpm

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Practice questions

Q1. Find the period and angular frequency of a 50 Hz supply.

Q2. v = 325 sin 314t. Find Vm, f, and v at t = 5 ms.

Q3. A 6-pole alternator runs at 1000 rpm. Find its frequency.

Q4. Why does the EMF reverse polarity every half cycle?